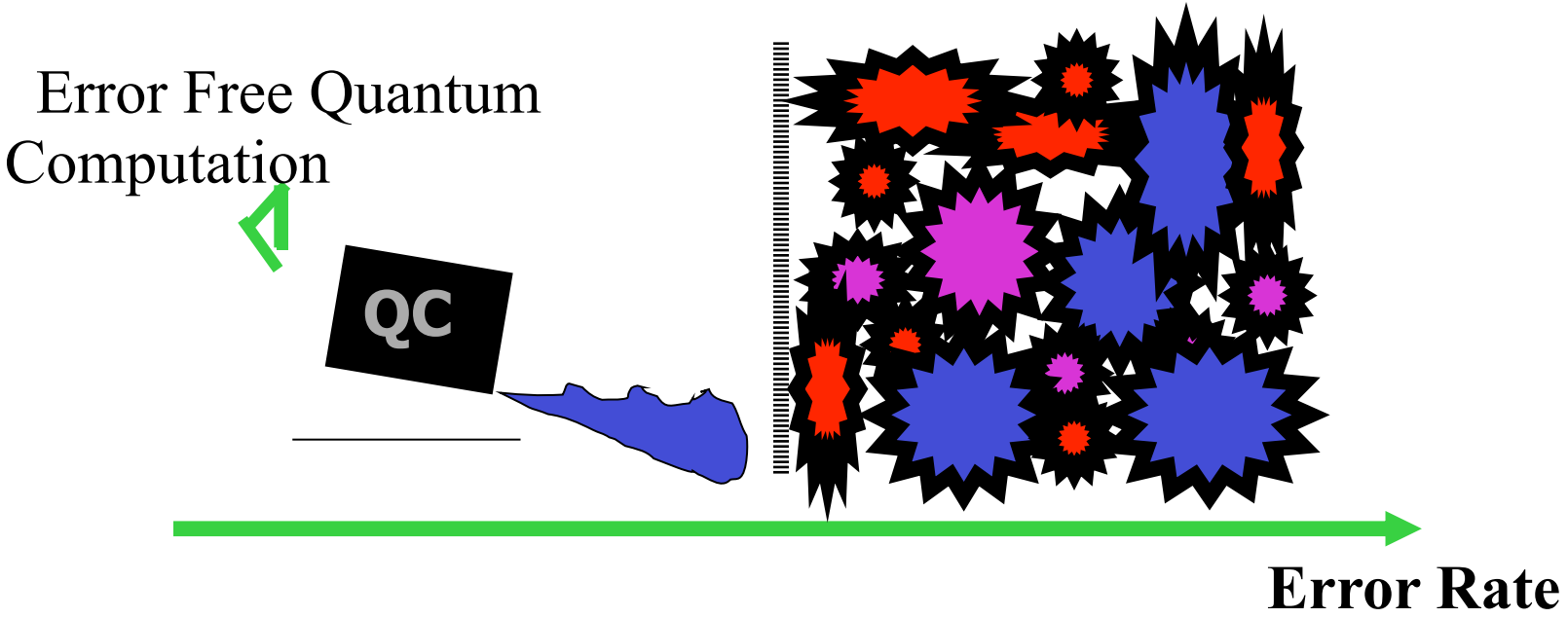
Fault-Tolerant Quantum Computation



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Mysterious Extra Slide

Please Do Not Read This

Quick Review

"The past exists only as recorded in the present."

- Operator Sum Representation (OSR) of open system evolution:

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger, \quad \sum_k \mathbf{A}_k^\dagger \mathbf{A}_k = \mathbf{I}$$

- Bit flip code overcomes no-cloning.
- Phase errors corrected by bit flip code in different basis.
- Shor code corrects single qubit errors.
- Quantum Error Correcting Code iff condition

$$\langle i | \mathbf{E}_a^\dagger \mathbf{E}_b | j \rangle = C_{ab} \delta_{ij}$$

- Errors can be "digitized".
- Argued for independent error model.

Old School

Bit flip code: $|0\rangle_L \to |000\rangle, |1\rangle_L \to |111\rangle$

$|\psi_C\rangle$ is in the codespace ($|\psi_C\rangle = \alpha|0\rangle_L + \beta|1\rangle_L$) iff

$$\mathbf{ZZI}|\psi_C\rangle = |\psi_C\rangle \quad \text{and} \quad \mathbf{IZZ}|\psi_C\rangle = |\psi_C\rangle$$

true iff first two qubits are $|00\rangle$ or $|11\rangle$

true iff last two qubits are $|00\rangle$ or $|11\rangle$

Codespace is defined as common $+1$ eigenvalue eigenspace of $\mathbf{S}_1 = \mathbf{ZZI}$ and $\mathbf{S}_2 = \mathbf{IZZ}$: $\mathbf{S}_\alpha|\psi_C\rangle = |\psi_C\rangle \forall \alpha$.

How do we see that a code satisfies the quantum error correcting condition:

$$\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = C_{ab}\delta_{ij}?$$

If $\exists \mathbf{S}_\alpha | \{\mathbf{S}_\alpha, \mathbf{E}_a^\dagger \mathbf{E}_b\} = 0$, then $\{\mathbf{A}, \mathbf{B}\} = \mathbf{AB} + \mathbf{BA}$

$$\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \langle i|\mathbf{E}_a^\dagger \mathbf{E}_b \mathbf{S}_\alpha|j\rangle = -\langle i|\mathbf{S}_\alpha \mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = -\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = 0$$

Code is for single qubit bit flip errors: $\mathbf{E}_a \in \{\mathbf{III}, \mathbf{XII}, \mathbf{IXI}, \mathbf{IIX}\}$

$$\mathbf{E}_a^\dagger \mathbf{E}_b \in \{\mathbf{III}, \mathbf{IIX}, \mathbf{IXI}, \mathbf{XII}, \mathbf{XXI}, \mathbf{XIX}, \mathbf{XXI}\}$$

Recall different Pauli operators anticommute: $\{\boldsymbol{\sigma}_\alpha, \boldsymbol{\sigma}_\beta\} = 2\delta_{\alpha,\beta}$:

$$\{\mathbf{X}, \mathbf{Y}\} = \{\mathbf{Y}, \mathbf{Z}\} = \{\mathbf{Z}, \mathbf{X}\} = 0$$

Let $\mathbf{P} = \mathbf{P}_1\mathbf{P}_2\cdots\mathbf{P}_n$, $\mathbf{Q} = \mathbf{Q}_1\mathbf{Q}_2\cdots\mathbf{Q}_n$ where $\mathbf{P}_i, \mathbf{Q}_j \in \{\mathbf{I}, \mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$.
Let $i_1, i_2, \ldots, i_k$ denote the locations where $\mathbf{P}$ and $\mathbf{Q}$ are both $\in \{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$ but differ.

If k is odd, then $\{\mathbf{P}, \mathbf{Q}\} = 0$.
If k is even, then $[\mathbf{P}, \mathbf{Q}] = 0$.

Example: $\{\mathbf{XXZZ}, \mathbf{YYXI}\} = 0 \quad [\mathbf{XXZ}, \mathbf{YZI}] = 0$

If $\mathbf{E}_a^\dagger \mathbf{E}_b \in \{\mathbf{IXI}, \mathbf{XII}, \mathbf{XIX}, \mathbf{IXX}\}$, then $\{\mathbf{ZZI}, \mathbf{E}_a^\dagger \mathbf{E}_b\} = 0$
If $\mathbf{E}_a^\dagger \mathbf{E}_b \in \{\mathbf{IIX}, \mathbf{IXI}, \mathbf{XIX}, \mathbf{XXI}\}$, then $\{\mathbf{IZZ}, \mathbf{E}_a^\dagger \mathbf{E}_b\} = 0$

$$\boxed{\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = 0, \quad \forall \mathbf{E}_a^\dagger \mathbf{E}_b \neq \mathbf{III}}$$

Pauli Group

The Pauli Group $\mathtt{P}_n$ on n qubits is the group of operators which are the tensor product of Pauli elements $\{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$ and the indentity $\mathbf{I}$ on all elements, plus a global phase $\pm 1, \pm i$

$$\mathbf{P} = i^p \mathbf{P}_1 \otimes \mathbf{P}_2 \otimes \cdots \otimes \mathbf{P}_n = \mathbf{P}_1 \mathbf{P}_2 \cdots \mathbf{P}_n, \quad \mathbf{P}_i \in \{\mathbf{I}, \mathbf{X}, \mathbf{Y}, \mathbf{Z}\}, p \in \{0, 1, 2, 3\}$$

Facts

- Any elements $\mathbf{P}, \mathbf{Q} \in \mathtt{P}_n$ either commute or anticommute, but not both

$$[\mathbf{P}, \mathbf{Q}] = 0 \quad \text{xor} \quad \{\mathbf{P}, \mathbf{Q}\} = 0$$

- Group members are traceless,

$$\mathrm{Tr}[\mathbf{P}] = 0, \ \forall \mathbf{P} \neq i^p \mathbf{I}$$

  and square to $\pm$ identity

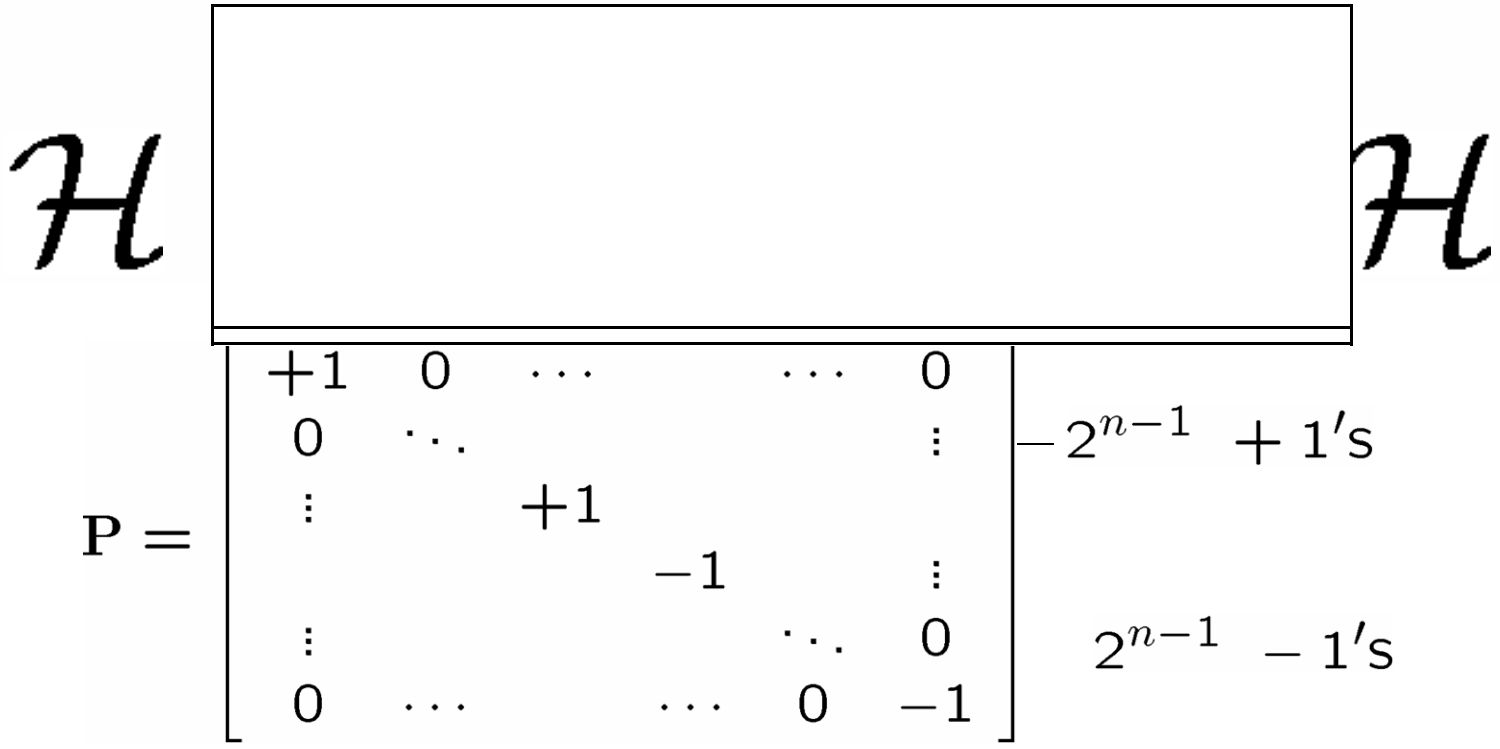
$$\mathbf{P}^2 = \pm \mathbf{I}$$

Mysterious Extra Slide II

Set Your Square Phasors For +1

Focus on the elements of $\mathtt{P}_n$ such that $\mathbf{P}^2 = \mathbf{I}$:

- Eigenvalues of such $\mathbf{P}$ are ± 1: $\mathbf{P}^2|\psi\rangle = \lambda^2|\psi\rangle = |\psi\rangle \rightarrow \lambda = \pm 1$.
- Degeneracies of $+1$ and -1 eigenvalues are 2^{n-1} for nonidentity $\mathbf{P}$:



The $\mathbf{P}$ divide the space in half depending on the ± 1 eigenvalue.

Abelian not Reptilian

A *Pauli Stabilizer Group* on n qubits is an Abelian Subgroup of the Pauli Groupon n qubits whose elements all square to identity.

$$\mathrm{S} = \{\mathbf{S}_\alpha \in \mathrm{P} | \mathbf{S}_\alpha^2 = \mathbf{I} \text{ and } [\mathbf{S}_\alpha, \mathbf{S}_\beta] = 0, \forall \alpha, \beta\}$$

Example: $\mathrm{S}_{ex} = \{\mathbf{IIII}, \mathbf{XXXX}, \mathbf{YYYY}, \mathbf{ZZZZ}\}$

The *generators* of a Pauli Stabilizer Group S is the smallest set whose elements can be multiplied to obtain the entire group [not unique].

Example: S_{ex} is generated by $\{\mathbf{XXXX}, \mathbf{ZZZZ}\}$

$$\mathrm{S}_{ex} = \langle \mathbf{XXXX}, \mathbf{ZZZZ} \rangle$$

Since group is abelian and square to identity, every element can be expressed as

$$\mathbf{S}_1^{\alpha_1} \mathbf{S}_2^{\alpha_2} \cdots \mathbf{S}_l^{\alpha_l}$$

where $\mathbf{S}_i$ generate the group and $\alpha_i \in \{0, 1\}$.

Example: $\mathbf{S} = (\mathbf{XXXX})^{\alpha_1} (\mathbf{ZZZZ})^{\alpha_2}, \ \forall \ \mathbf{S} \in \mathrm{S}_{ex}$

Pauli Stabilizer Code

A Pauli Stabilizer Code S for a Pauli Stabilizer Group $\mathbb{S}$ is defined as the common $+1$ eigenvalue eigenspace of the elements of $\mathbb{S}$:

$$|\psi_C\rangle \in S \text{ iff } \mathbf{S}_\alpha|\psi_C\rangle = |\psi_C\rangle, \ \forall \mathbf{S}_\alpha \in \mathbb{S}$$

Since

$$\mathbf{S}(\alpha_1, \alpha_2, \ldots, \alpha_k) = \mathbf{S}_1^{\alpha_1}\mathbf{S}_2^{\alpha_2}\cdots\mathbf{S}_k^{\alpha_k}$$

we need only check generators:

$$|\psi_C\rangle \in S \text{ iff } \mathbf{S}_\alpha|\psi_C\rangle = |\psi_C\rangle, \ \forall \mathbf{S}_\alpha \in \langle \mathbb{S} \rangle$$

Truth By Example

Example: $\mathsf{S}_{ex} = \langle \mathbf{XXXX}, \mathbf{ZZZZ}, \mathbf{XXII} \rangle$

$\mathbf{ZZZZ}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ computational basis states are even parity

$$|\psi_C\rangle \in \text{Span}\,[|0000\rangle, |0011\rangle, |0110\rangle, |0101\rangle, |1100\rangle, |1010\rangle, |1001\rangle, |1111\rangle]$$

$\mathbf{XXXX}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ use projector $\frac{1}{2}(\mathbf{IIII} + \mathbf{XXXX})$

$$|\psi_C\rangle \in \text{Span}\left[\frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle), \frac{1}{\sqrt{2}}(|0110\rangle + |1001\rangle), \frac{1}{\sqrt{2}}(|1100\rangle + |0011\rangle), \frac{1}{\sqrt{2}}(|1010\rangle + |0101\rangle)\right]$$

$\mathbf{XXII}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ use projector $\frac{1}{2}(\mathbf{IIII} + \mathbf{XXII})$

$$|\psi_C\rangle \in \text{Span}\left[\frac{1}{2}(|0000\rangle + |1111\rangle + |1100\rangle + |0011\rangle), \frac{1}{2}(|0110\rangle + |0110\rangle + |1010\rangle + |0101\rangle)\right]$$

Sizing Up The Code

Let $\mathtt{S}$ generated by $\mathbf{G}_1, \mathbf{G}_2, \ldots, \mathbf{G}_l$. Define the projectors onto the $+1$ eigenspace:

$$\mathbf{P}_i = \frac{1}{2}(\mathbf{I} + \mathbf{G}_i)$$

$$\mathbf{P}_i\mathbf{P}_j = \mathbf{P}_j\mathbf{P}_i, \quad \mathbf{P}_i^2 = \mathbf{P}_i, \quad \mathrm{Tr}[\mathbf{P}_i] = 2^{n-1}$$

Projector onto the codespace is

$$\mathbf{P} = \mathbf{P}_1\mathbf{P}_2\cdots\mathbf{P}_l$$

$$\mathbf{P}^2 = \mathbf{P}, \quad \dim C = \mathrm{Tr}[\mathbf{P}]$$

$$\mathrm{Tr}\left[\mathbf{P}_1\mathbf{P}_2\right] = \frac{1}{4}\mathrm{Tr}\left[\mathbf{I} + \mathbf{G}_1 + \mathbf{G}_2 + \mathbf{G}_1\mathbf{G}_2\right] = 2^{n-2}$$

Not identity (generators!)

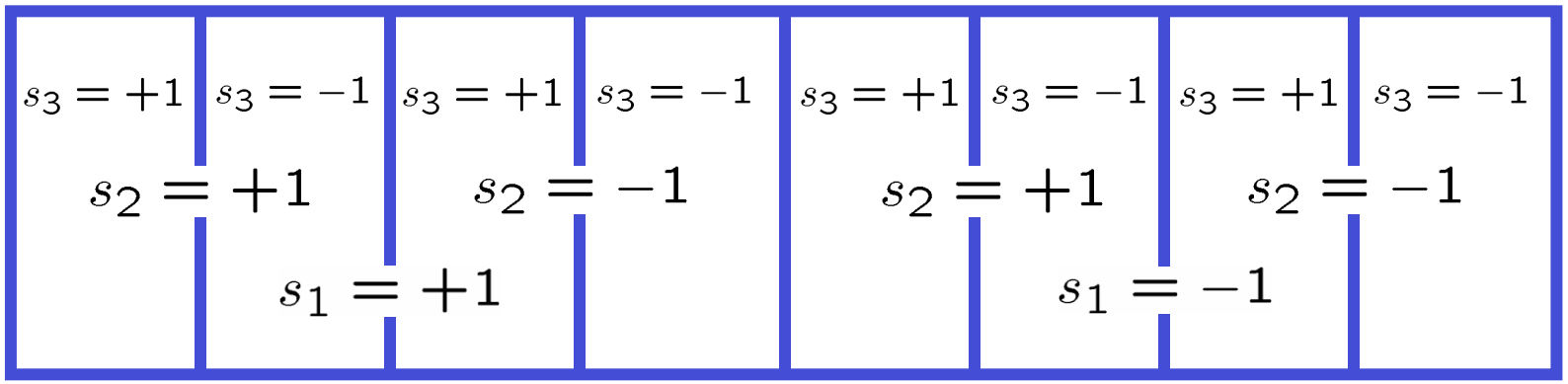
$$\mathbf{P} = \frac{1}{2^l} \sum_{\alpha_1,\alpha_2,\ldots,\alpha_l \in \{0,1\}} \mathbf{G}_1^{\alpha_1}\mathbf{G}_2^{\alpha_2}\cdots\mathbf{G}_l^{\alpha_l} \Rightarrow \mathrm{Tr}[\mathbf{P}] = 2^{n-l}$$

A stabilizer code generated by k elements has dimension 2^{n-k} (encodes $n-k$ qubits)

Stabilizer Subspaces

$\mathbf{G}_i|\psi\rangle = |\psi\rangle,\ \forall i$ defines a 2^{n-l} dimensional subspace.

Every unique $s_i = \pm 1$ defines, via $\mathbf{G}_i|\psi\rangle = s_i|\psi\rangle,\ \forall i$, a *uniques* subspace.



2^n dimensional Hilbert space

s_1	s_2	
$+1$	$+1$	2^{n-2} dimensional codespace
$+1$	-1	2^{n-2} dimensional codespace
-1	$+1$	2^{n-2} dimensional codespace
-1	-1	2^{n-2} dimensional codespace

$|s_1, s_2, \ldots, s_k, m\rangle$, m labels 2^{n-l} subspace

Stabilizer Nerror

Stabilizer $\mathbb{S}$ with generators $\mathbf{G}_1, \ldots, \mathbf{G}_k$ stabilizes a code of $n-l$ qubits. What errors does this code correct?

Suppose $\mathbf{E}$ anticommutes with *at least* one $\mathbf{G}_i$,

$$\{\mathbf{E}, \mathbf{G}_i\} = 0 \quad \mathbf{E}\mathbf{G}_i = -\mathbf{G}_i\mathbf{E}$$

then as before, for stabilized codewords,

$$\langle i|\mathbf{E}|j\rangle = \langle i|\mathbf{E}\mathbf{G}_i|j\rangle = -\langle i|\mathbf{G}_i\mathbf{E}|j\rangle = -\langle i|\mathbf{G}|j\rangle = 0$$

If $\mathbf{E}$ is in the stabilizer:

$$\langle i|\mathbf{E}|j\rangle = \langle i|j\rangle = \delta_{ij}$$

Given a set of errors $\mathbf{E}_a$, a stabilizer code S corrects these errors if the products $\mathbf{E}_a^\dagger \mathbf{E}_b$ either

(i) anticommute with at least one stabilizer generator

(ii) are in the stabilzer group

Surfing The Subspaces

If $\mathbf{E}_a$ anticommutes with $\mathbf{G}_i$,

$$\mathbf{G}_i(\mathbf{E}_a|s_1,\ldots,s_l,m\rangle) = -s_i(\mathbf{E}_a|s_1,\ldots,s_l,m\rangle)$$

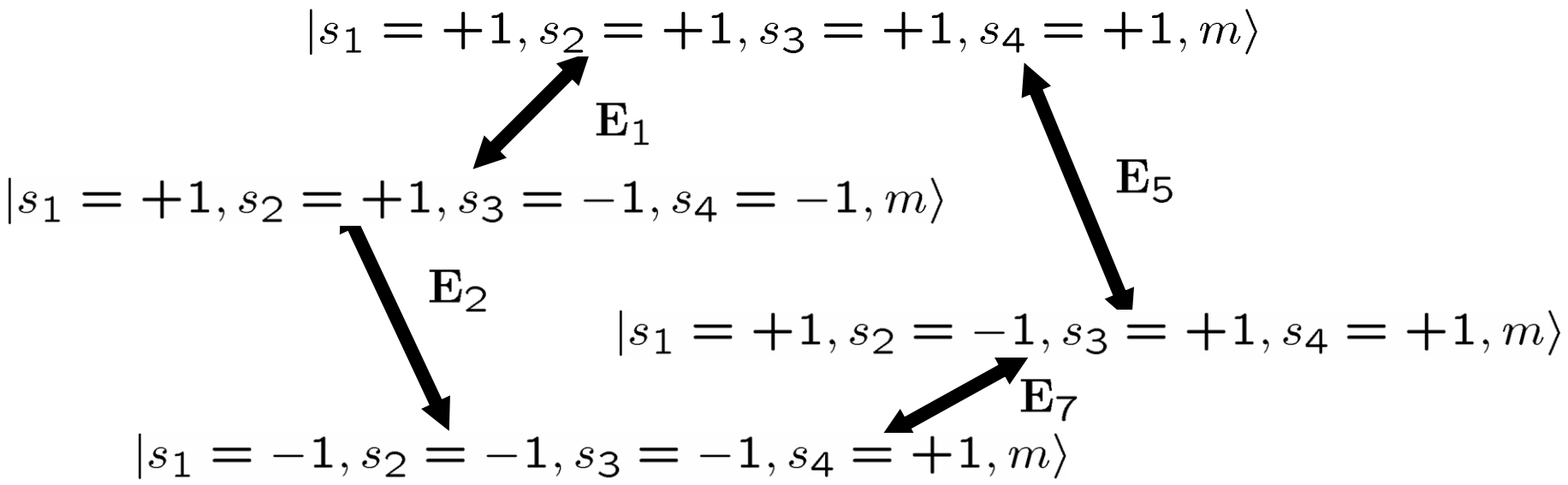
$$|s_1,s_2,s_3,\ldots,s_l,m\rangle$$

$\mathbf{E}_a$

$$|\bar{s}_1,\bar{s}_2,\bar{s}_3,\ldots,\bar{s}_l,m\rangle$$

$\bar{s}_i = s_i$ if $[\mathbf{G}_i,\mathbf{E}_a] = 0$

$\bar{s}_i = -s_i$ if $\{\mathbf{G}_i,\mathbf{E}_a\} = 0$



Unsurfing The Subspaces

Measuring the generators $\mathbf{G}_i$ can be simultaneously performed (they commute!).

Measuring the generators $\mathbf{G}_i$ determines which $s_1, s_2, \ldots, s_k$ subspace the state is in.

Once the s_i are determined, returning to the codespace is simply rotating back to all $s_i = +1$?

$$|s_1 = +1, s_2 = +1, s_3 = +1, s_4 = +1, m\rangle$$



Error

$$|s_1 = +1, s_2 = -1, s_3 = -1, s_4 = +1, m\rangle$$



Fix ⟵ Correct fix?

$$|s_1 = +1, s_2 = +1, s_3 = +1, s_4 = +1, m\rangle$$

Well yes, from iff condition

Examples

$\langle \mathbb{S} \rangle = \{\mathbf{ZZI}, \mathbf{IZZ}\}$ corrects single bit flip errors $\{\mathbf{E}_a\} = \{\mathbf{XII}, \mathbf{IXI}, \mathbf{IIX}\}$.

$\langle \mathbb{S} \rangle = \{\mathbf{XXI}, \mathbf{IXX}\}$ corrects single phase errors $\{\mathbf{E}_a\} = \{\mathbf{ZII}, \mathbf{IZI}, \mathbf{IIZ}\}$.

$$\langle \mathbb{S} \rangle = \{\mathbf{ZZI|III|III}, \mathbf{IZZ|III|III}, \mathbf{III|ZZI|III}, \mathbf{III|IZZ|III}, \mathbf{III|III|ZZI}, \mathbf{III|III|IZZ}, \mathbf{XXX|XXX|III}, \mathbf{III|XXX|XXX}\}$$

Shor code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

$\langle \mathbb{S} \rangle = \{\mathbf{XZZXI}, \mathbf{IXZZX}, \mathbf{XIXZZ}, \mathbf{ZXIXZ}\}$

Five bit code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

All encode one qubit.

$\langle \mathbb{S} \rangle = \{\mathbf{ZZZZZZZZ}, \mathbf{XXXXXXXX}, \mathbf{IXIXYZYZ}, \mathbf{IXZYIXZY}, \mathbf{IYXZIYXZ}\}$

8 qubit code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

. . . but encodes 3 qubits!

Quantum Hamming Bound

Suppose all errors anticommute with at least one stabilizer generator (non-degenerate code).

n qubit code which corrects t or less errors and encodes k qubits.

If j error occur, $\binom{n}{j}$ ways to place these errors.

Each of the j errors can be one of 3 errors.

Total number of errors $\sum_{j=0}^{t} \binom{n}{j} 3^j$

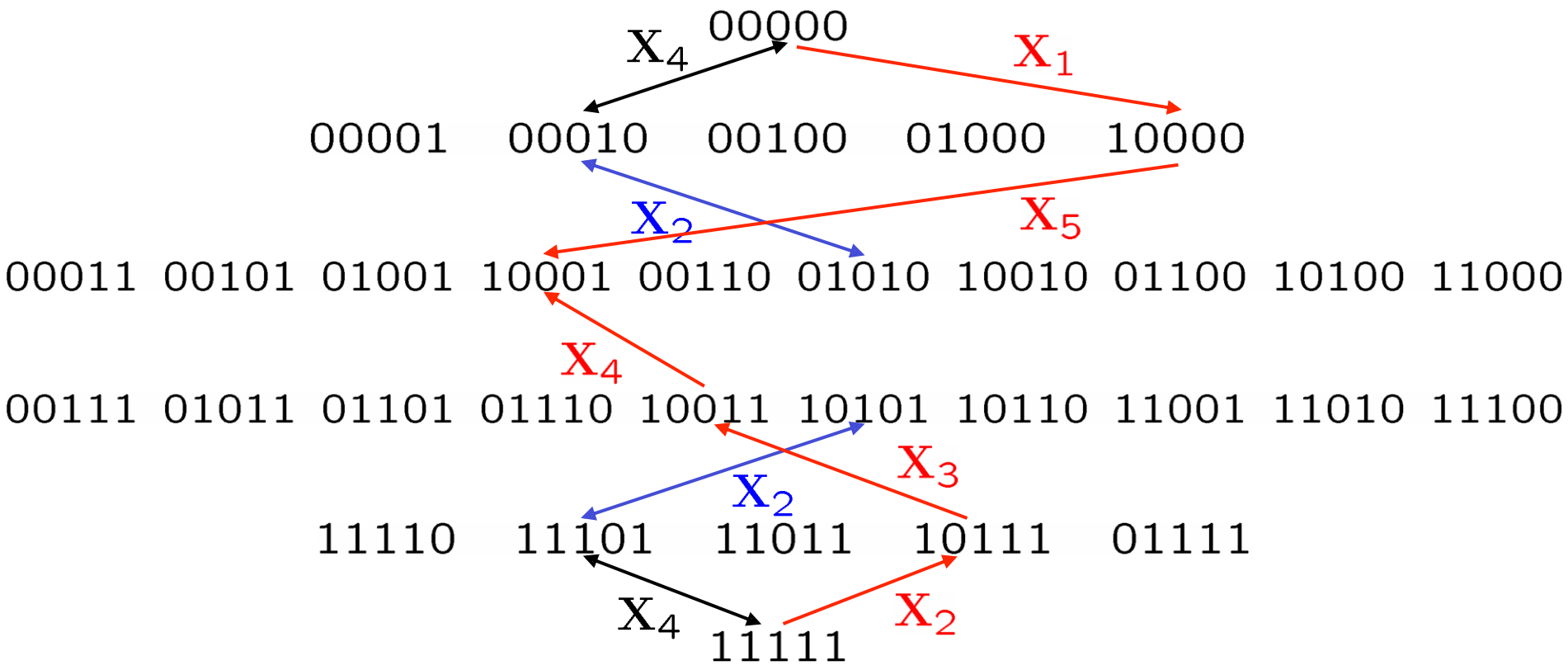
Each error gets its own subspace of dimension 2^k and all this must fit in the n qubit space

$$\sum_{j=0}^{t} \binom{n}{j} 3^j 2^k \leq 2^n$$

$$t = k = 1 \Rightarrow 2(1 + 3n) \leq 2^n \Rightarrow n \geq 5$$

Research Problem: BEAT THE QUANTUM HAMMING BOUND

Detect Distinguishable Correct



Weight of Pauli group member = # non-identity tensor elements

Weight $(\mathbf{XYIZ})$=3

Smallest Weight for a code which takes codeword A to codeword $B = d =$*distance*. A code which corrects t errormust be distance $2t + 1$.

k encoded qubits

n qubits → $[n, k, d]$ code ← distance d

Pauli Want a Qubit?

Return to our old friend the bit flip code: $\mathbf{S}_1 = \mathbf{ZZI}, \mathbf{S}_2 = \mathbf{IZZ}$

$$C = \text{Span}[|000\rangle, |111\rangle]$$

How do we perform encoded opeator on code?

Example: encoded "$\mathbf{X}$": $|000\rangle \Leftrightarrow |111\rangle$

$$\begin{array}{lllllllll} \mathbf{XXX}|000\rangle & = & |111\rangle & -\mathbf{XYY}|000\rangle & = & |111\rangle & -\mathbf{YXY}|000\rangle & = & |111\rangle \\ \mathbf{XXX}|111\rangle & = & |000\rangle & -\mathbf{XYY}|111\rangle & = & |000\rangle & -\mathbf{YXY}|111\rangle & = & |000\rangle \end{array}$$

What is interesting about these operators $(\mathbf{XXX}, \mathbf{XYY}, \mathbf{YXY})$?

They commute with all stabilizer group elements . . .

. . . but are not themselves elements of the stabilizer group.

$$\mathrm{S} = \{\mathbf{III}, \mathbf{ZZI}, \mathbf{IZZ}, \mathbf{ZIZ}\}$$

$(\mathbf{XXX})\mathbf{S}$, $\mathbf{S} \in \mathrm{S}$ acts on code same as $\mathbf{XXX}$

Modulo the stabilizer $\mathbf{S}$

$\mathbf{S}_1 = \mathbf{ZZI}, \mathbf{S}_2 = \mathbf{IZZ}$

Logical opeators: $\bar{\mathbf{X}} = \mathbf{XXX}$, $\bar{\mathbf{Y}} = \mathbf{YXX}$, $\bar{\mathbf{Z}} = \mathbf{ZII}$

$$\mathbf{XXX}|000\rangle = |111\rangle \qquad \mathbf{ZII}|000\rangle = |000\rangle$$
$$\mathbf{XXX}|111\rangle = |000\rangle \qquad \mathbf{ZII}|111\rangle = -|111\rangle$$

Modulo the stabilizer $\mathbf{S}$

Feel free to randomly insert "Modulo the stabilizer $\mathbf{S}$" into my sentence

Logical opeators used to define "encoded" computational basis ± 1 eigenvalues of $\bar{Z}$.

Logical Operators for a Stabilizer Code

Given a stabilizer group $\mathbb{S}$, the set of Pauli operators which commute with all of the elements of $\mathbb{S}$, $\{\mathbf{L}|[\mathbf{L}, \mathbf{S}], \forall \mathbf{S} \in \mathbb{S}\}$, for the logical operators (MTS) for the stabilizer code.

$\mathbf{S}(\mathbf{L}|\psi_C\rangle) = \mathbf{L}(\mathbf{S}|\psi_C\rangle) = (\mathbf{L}|\psi_C\rangle$: $\mathbf{L}$ preserves codespace.

Pauli Wants Lots of Qubits

Consider the code with a single stabilizer element on 3 qubits:

$$\mathbf{S}_1 = \mathbf{XXX}$$

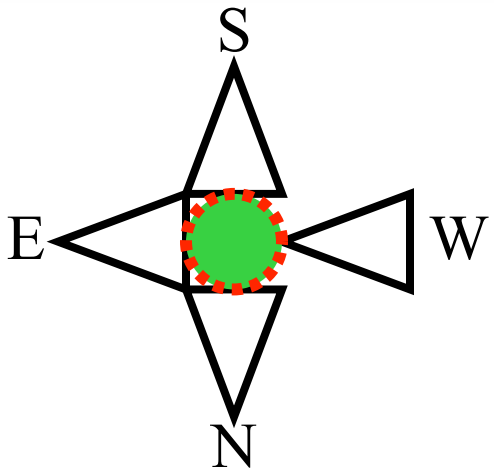
Should encode 2 qubits: what are the logical operators?

$$\begin{aligned} \bar{\mathbf{X}}_1 &= \mathbf{XII} \quad \bar{\mathbf{Z}}_1 = \mathbf{ZZI} \\ \bar{\mathbf{X}}_2 &= \mathbf{IIX} \quad \bar{\mathbf{Z}}_2 = \mathbf{IZZ} \end{aligned}$$

$$\bar{\mathbf{X}}_1\bar{\mathbf{Z}}_1 = -i\bar{\mathbf{Y}}_1 \qquad \bar{\mathbf{X}}_2\bar{\mathbf{Z}}_2 = -i\bar{\mathbf{Y}}_2 \qquad [\bar{\mathbf{P}}_1, \bar{\mathbf{P}}_2] = 0$$

But what is qubit 1 and qubit 2? Basis change renders notion of subsystem irrelevant: $\mathbf{U}^\dagger \mathbf{P}_i \mathbf{U}$

$$\begin{aligned} \bar{\mathbf{X}}_1 &= \mathbf{XIX} \quad \bar{\mathbf{Z}}_1 = \mathbf{ZZI} \\ \bar{\mathbf{X}}_2 &= \mathbf{IIX} \quad \bar{\mathbf{Z}}_2 = \mathbf{ZIZ} \end{aligned}$$



Who says north is up?

Real Life (I'm Plato) Example

The Calderbank-Shor-Steane $[7, 1, 3]$ code C_{CSS}

$$\begin{aligned}\langle \mathrm{S} \rangle \;=\; &\{\mathbf{XXXXIII}, \mathbf{XXIIXXI}, \mathbf{XIXIXIX},\\ &\mathbf{ZZZZIII}, \mathbf{ZZIIZZI}, \mathbf{ZIZIZIZ}\}\end{aligned}$$

7 physical qubits, 1 encoded qubit, corrects all single qubit errors.

Logical Operators are

$$\bar{\mathbf{X}} = \mathbf{XXXXXXX} \quad \bar{\mathbf{Z}} = \mathbf{ZZZZZZZ}$$

(Nobody ever rights $\bar{\mathbf{Y}}$ because it is just $\bar{\mathbf{Y}} = i\bar{\mathbf{X}}\bar{\mathbf{Z}}$)

Clifford the Dog

Maintain codespace if

$$\mathbf{O}|\psi_C\rangle = \mathbf{O}\mathbf{S}_\alpha|\psi_C\rangle = \mathbf{S}_\beta\mathbf{O}|\psi_C\rangle, \ \forall\alpha,\beta$$

Operators $\mathbf{O}$ which permute the stabilizer elements:

$$\mathbf{O}\mathbf{S}_\alpha = \mathbf{S}_\beta\mathbf{O} \Rightarrow \mathbf{O}^\dagger\mathbf{S}_\beta\mathbf{O} = \mathbf{S}_\alpha$$

Logical operators are example of such $\mathbf{O}$ for which $\alpha = \beta$.

Example: $\mathbb{S} = \langle \mathbf{XXXX}, \mathbf{ZZZZ} \rangle$

$$(\mathbf{HHHH})\mathbf{XXXX}(\mathbf{HHHH}) = \mathbf{ZZZZ}$$
$$(\mathbf{HHHH})\mathbf{YYYY}(\mathbf{HHHH}) = \mathbf{YYYY}$$

Example: C_{CSS} $\bar{\mathbf{X}} = \mathbf{XXXXXXX}$ $\bar{\mathbf{Z}} = \mathbf{ZZZZZZZ}$

$$(\mathbf{HHHHHHH})\bar{\mathbf{X}}(\mathbf{HHHHHHH}) = \bar{\mathbf{Z}}$$
$$(\mathbf{HHHHHHH})\bar{\mathbf{Y}}(\mathbf{HHHHHHH}) = -\bar{\mathbf{Y}}$$

Conjoining Subsystems



encoded qubits with stabilizer $\langle \mathbf{S}_1, \ldots, \mathbf{S}_l \rangle$

encoded qubits with stabilizer $\langle \mathbf{T}_1, \ldots, \mathbf{T}_m \rangle$

Conjoined stabilizer group $\mathtt{S} \otimes \mathtt{T} = \langle \mathbf{S}_1 \otimes \mathbf{T}_1, \mathbf{S}_1 \otimes \mathbf{T}_2, \ldots, \mathbf{S}_l \otimes \mathbf{T}_m \rangle$

Is state of conjoined system $|\psi_C\rangle$ (not necessarily seperable) within the code?

Use conjoined stabilizer group $\mathbf{S}_\alpha \otimes \mathbf{T}_\beta |\psi_C\rangle = |\psi_C\rangle$

Stabilizer In Review

- Stabilizer group $\mathbb{S}$ is an abelian subgroup of the Pauli group with l generators $\mathbf{S}_\alpha$.

- The code corresponding to the stabilizer group $\mathbb{S}$ is the subspace spanned by the states which are stabilized by all of the stabilizer group operators $\mathbf{S}|\psi_C\rangle = |\psi_C\rangle$.

- The code corrects the set of errors $\{\mathbf{E}_a\}$ if $\mathbf{E}_a^\dagger \mathbf{E}_b$ anticommutes which at least one stabilizer element or $\mathbf{E}_a^\dagger \mathbf{E}_b$ is in the stabilizer.

- Logical operators for the encoded qubits are Pauli operators which commute with all the stabilizer elements but which are not themselves stabilizer elements.

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Decoherence Free Subspaces

Two qubits, suppose error is $\mathbf{XX}$'s:

$$\boldsymbol{\rho}_{out} = (1-p)\boldsymbol{\rho}_{in} + p\mathbf{XX}\boldsymbol{\rho}_{in}\mathbf{XX}$$

$$|0_L\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad |1_L\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

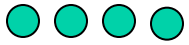
$$\mathbf{XX}|0_L\rangle = |0_L\rangle, \mathbf{XX}|1_L\rangle = |1_L\rangle$$
$$\Rightarrow \mathbf{XX}(\alpha|0_L\rangle + \beta|1_L\rangle) = (\alpha|0_L\rangle + \beta|1_L\rangle)$$

Encode into the subspace spanned by $|0_L\rangle$ and $|1_L\rangle$: $\boldsymbol{\rho}_{out} = \boldsymbol{\rho}_{in}$

Symmetry of error process can be used to find quiet corner of Hilbert space.

Collective Decoherence

spins coupled to bath field, N qubits $s_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $s_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $s_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$



system-bath Hamiltonian

Bath mode with a wavelength much longer than the inner qubit spacing

$$H_I = \sum_{i=1}^{n} \sum_{\alpha=\{x,y,z\}} c_{\alpha,i} \underbrace{s_\alpha^{(i)}}_{\text{system operator acting on } i\text{th qubit}} \otimes \underbrace{b_{\alpha,i}}_{\text{bath operator}} \Leftrightarrow H_I = \sum_{\alpha=\{x,y,z\}} \sum_{i=1}^{n} c_\alpha s_\alpha^{(i)} \otimes b_\alpha = \sum_{\alpha=\{x,y,z\}} S_\alpha^n \otimes B_\alpha$$

$c_{\alpha,i} = c_\alpha$

$$s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I$$

bath couples identically to each qubit

collective spin operators:

$$S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)}$$

$$[s_\alpha, s_\beta] = i\varepsilon_{\alpha\beta\gamma} s_\gamma$$

su(2) Lie algebras

$$[S_\alpha^n, S_\beta^n] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$

Just Angular Momentum Addition

Collective su(2) for n=2 qubits:

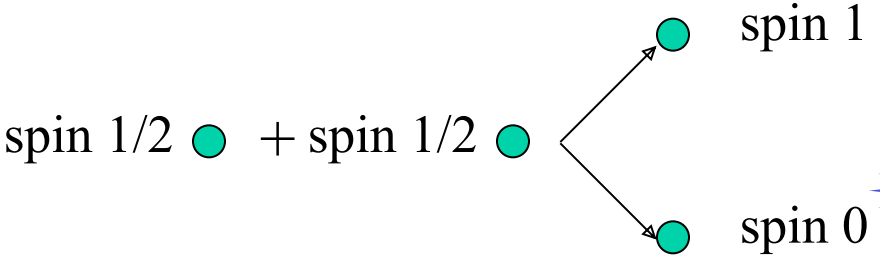
$$S_x^2 = \frac{1}{\sqrt{2}}\left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_y^2 = \frac{1}{\sqrt{2}}\left(\begin{array}{ccc|c} 0 & -i & 0 & 0 \\ i & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \qquad S_z^2 = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

triplet states $\begin{cases} |J=1, m_J=1\rangle = |00\rangle \\ |J=1, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \\ |J=1, m_J=-1\rangle = |11\rangle \end{cases}$

singlet state $\left\{ |J=0, m_J=0\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) \right.$

red is the 3 dimensional irreducible representation of su(2) (J=1)

blue is the 1 dimensional irreducible representation of su(2) (J=0)



simple addition of angular momentum…

nothing to see here, but:

Degeneracy

Something special happens with n=3:

•spin-1 (triplet) plus spin-1/2 can produce either a spin-1/2 or a spin-3/2.

•spin-0 (singlet) plus spin-1/2 can produce spin 1/2.

$$S_x^3 = \left(\begin{array}{cccc|cc|cc} 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{\frac{3}{2}} & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \end{array}\right)$$

$$\left\{\begin{array}{l} \left|J=\frac{3}{2},m_J=\frac{3}{2}\right\rangle = \left|000\right\rangle \\ \left|J=\frac{3}{2},m_J=\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(\left|001\right\rangle+\left|010\right\rangle+\left|100\right\rangle\right) \\ \left|J=\frac{3}{2},m_J=-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(\left|011\right\rangle+\left|101\right\rangle+\left|110\right\rangle\right) \\ \left|J=\frac{3}{2},m_J=-\frac{3}{2}\right\rangle = \left|111\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\frac{1}{2},m_J=\frac{1}{2}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|0\right\rangle \\ \left|J=\frac{1}{2},m_J=-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|01\right\rangle-\left|10\right\rangle\right)\left|1\right\rangle \end{array}\right.$$

$$\left\{\begin{array}{l} \left|J=\frac{1}{2},m_J=\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|00\right\rangle\left|1\right\rangle-\sqrt{\frac{1}{3}}\left\{\frac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|0\right\rangle \\ \left|J=\frac{1}{2},m_J=-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|11\right\rangle\left|0\right\rangle-\sqrt{\frac{1}{3}}\left\{\frac{1}{\sqrt{2}}\left(\left|01\right\rangle+\left|10\right\rangle\right)\right\}\left|1\right\rangle \end{array}\right.$$

These spin-1/2 states come from combining a spin-0 two qubit state and a spin-1/2 single qubit state

These spin-1/2 states come from combining a spin-1 two qubit state and a spin-1/2 single qubit state

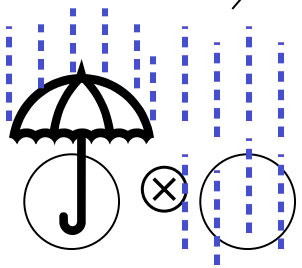
There is a **DEGENERACY** in the action of the 3 qubit collective su(2) operation.

Degeneracy Implies Safety

$$S_\alpha^3 = \left(\begin{array}{c|c:c} s_\alpha^4 & 0 & 0 \\ \hline 0 & s_\alpha^2 & 0 \\ \hdashline 0 & 0 & s_\alpha^2 \end{array}\right) = \left(\begin{array}{c|c} s_\alpha^4 & 0 \\ \hline 0 & I \otimes s_\alpha^2 \end{array}\right)$$

where $s_\alpha^k \equiv$ {the k-dimensional irreducible representation of su(2)}

$$S_\alpha^3 \rightarrow s_\alpha^4 \oplus \left(I_2 \otimes s_\alpha^2\right)$$



protected qubit unprotected qubit

Information encoded into the **DEGENERACY** is **immune** to collective decoherence

(KLV) Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Classical Example

clumsy hands channel: both coins get flipped with probability p

$|HH\rangle$
$|HT\rangle$
$|TH\rangle$
$|TT\rangle$



(1-p)



+p



$|$coin 1, coin 2 $\rangle$

Information can be immune to this channel if

encode into the parity

$|$parity,coin 1$\rangle$

$|1H\rangle=$



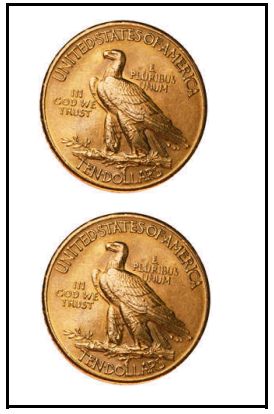
$|1T\rangle=$



$|0H\rangle=$



$|0T\rangle=$



$|$parity$\rangle=|0\rangle$ if coins same, $|$parity$\rangle=|1\rangle$ if coins differ

Decoherence Free Subsystems

General system-bath Hamiltonian $H_{SB} = \sum_{\alpha} A_\alpha \otimes B_\alpha$

A_α: system operators; B_α: linearly independent bath operators

A_α form an algebra A (add, multiply them) which is decomposable into subsystems:

$$\mathrm{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

M_{d_J}: linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator

n_J is the degeneracy of the Jth irrep

Quantum information encoded into a single degenerate component subsystem will not decohere! Decoherence-Free Subsystems (DFSs)

"The universe is an enormous direct product of representations of symmetry groups" - Steve Weinberg

<u>for open system becomes</u>

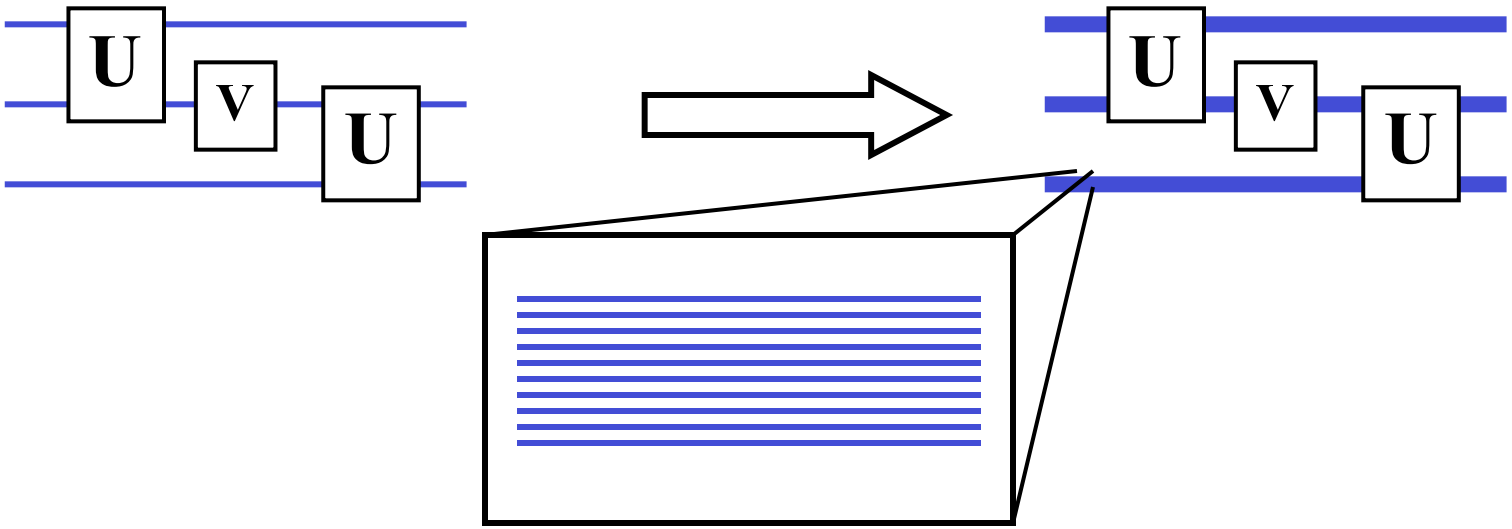
"The universe is an enormous direct product of full matrix algebras" - not Steve Weinberg

Fault-Tolerant Quantum Computation

Goal: robust quantum computation in presence of all reasonable imperfections.

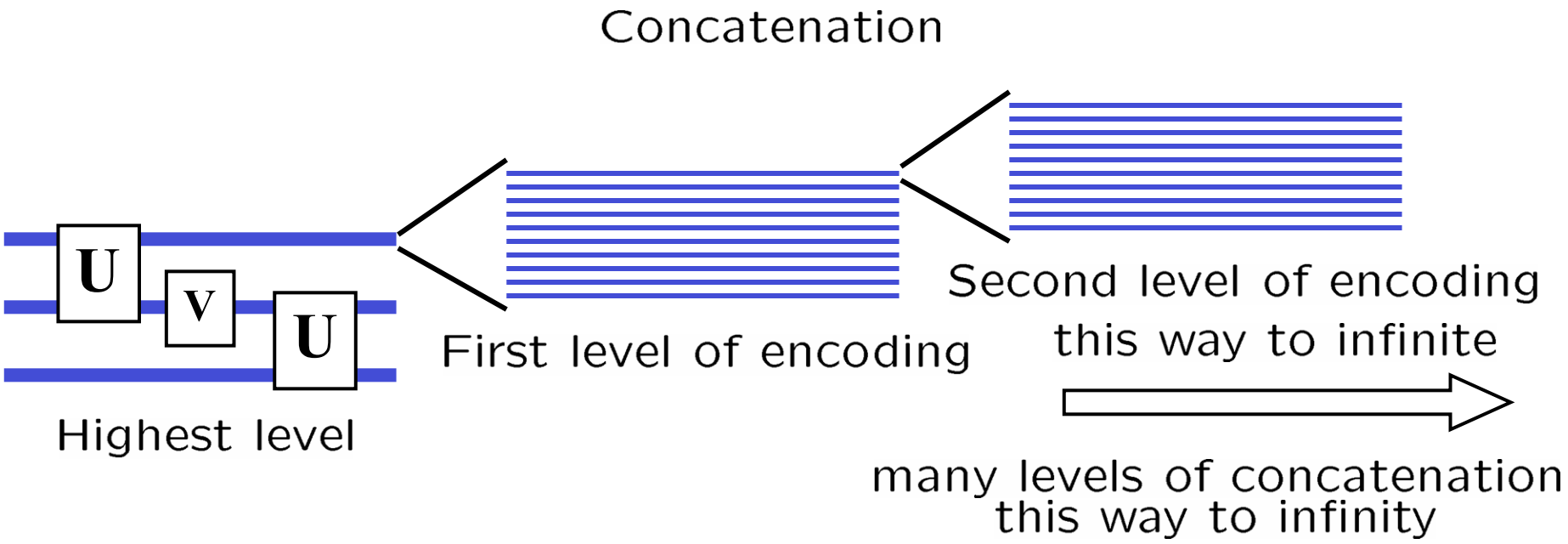
Decoherence, imperfect gates, imperfect measurements, imperfect preparation

Idea 1: Always work with quantum error corrected qubits



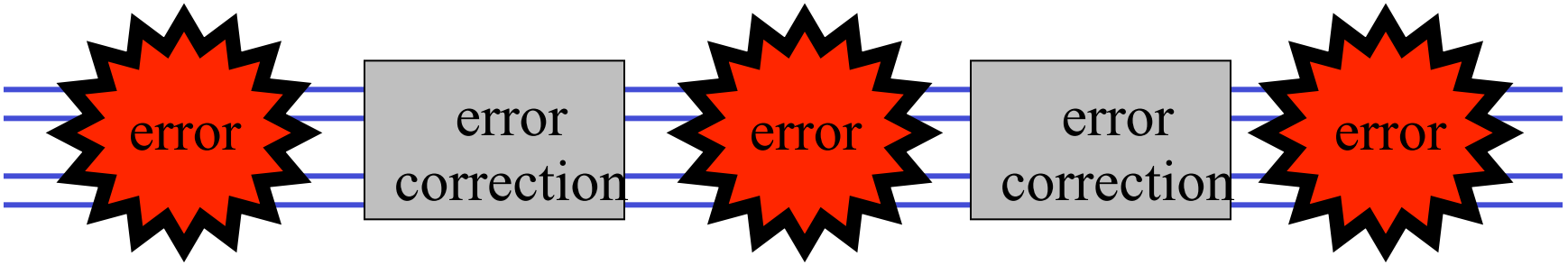
Perfection Through Concatenation

Idea 2: Error correct these encoded qubits to fix 'errors'



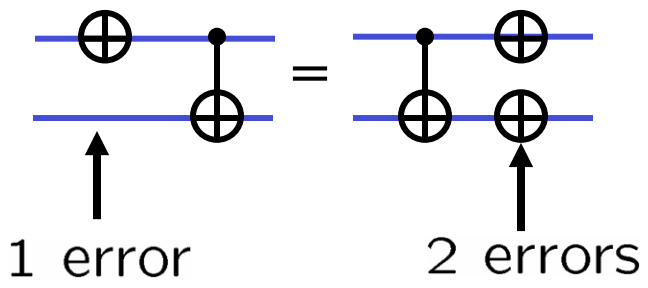
Concatenation allows toleration of more and more errors.

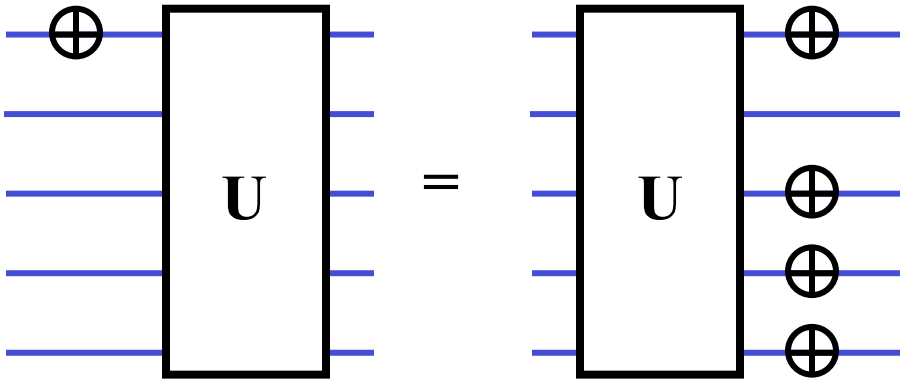
'probability of error' goes to zero like a^{-k}, k concatenation levels



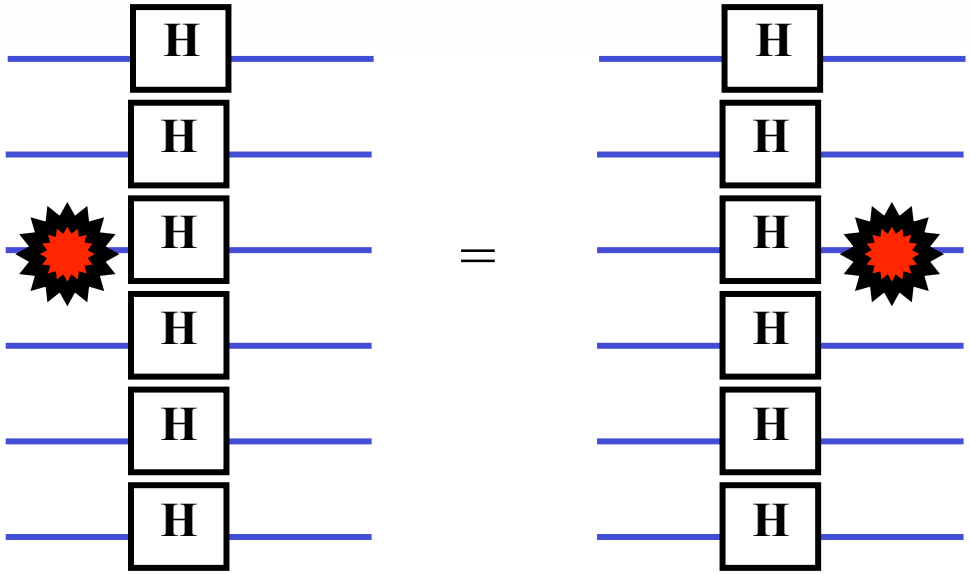
A Plague of Errors

Errors spread:





Idea 3: Design gates, preparation, measurement, to stop this plague



Bitwise operations do not spread errors plague

Be Careful Where You Rotate

Our old budy: $\mathbf{S}_1 = \mathbf{ZZI}, \mathbf{S}_2 = \mathbf{IZZ}$, $\bar{\mathbf{X}} = \mathbf{XXX}, \bar{\mathbf{Z}} = \mathbf{ZII}$

Two ways to get $\mathbf{U} = -i\mathbf{XXX}$

$\mathbf{U} = \exp[-i\mathbf{H}t]$, $\mathbf{H}$ is the Hamiltonian, t is time

$$-i\mathbf{XXX} = \exp[-i\mathbf{XXX}t]|_{t=\frac{\pi}{2}}$$

$$-i\mathbf{XXX} = \exp[-i(\mathbf{XII} + \mathbf{IXI} + \mathbf{IIX})t]|_{t=\frac{\pi}{2}}$$

$$\exp[-i\mathbf{H}t] = \cos(t)\mathbf{I} - i\sin(t)\mathbf{H} \text{ if } \mathbf{H}^2 = \mathbf{I}$$

What happens if you mess up timing by Δt?

$$\exp[-i\mathbf{XXX}(t+\Delta t)] = -i\mathbf{XXX}\,[\cos(\Delta t)\mathbf{I} - i\sin(\Delta t)\mathbf{XXX}]$$

$$\exp[-i(\mathbf{XII} + \mathbf{IXI} + \mathbf{IIX})(t+\Delta t)] = -i\mathbf{XXX}\,[(\cos(\Delta t)\mathbf{I} - i\sin(\Delta t)\mathbf{XII})$$
$$(\cos(\Delta t)\mathbf{I} - i\sin(\Delta t)\mathbf{IXI})(\cos(\Delta t)\mathbf{I} - i\sin(\Delta t)\mathbf{IIX})]$$

$$\exp[-i\mathbf{XXX}(t+\Delta t)] \approx -i\mathbf{XXX}\,[\mathbf{I} - i\Delta t\mathbf{XXX}]$$

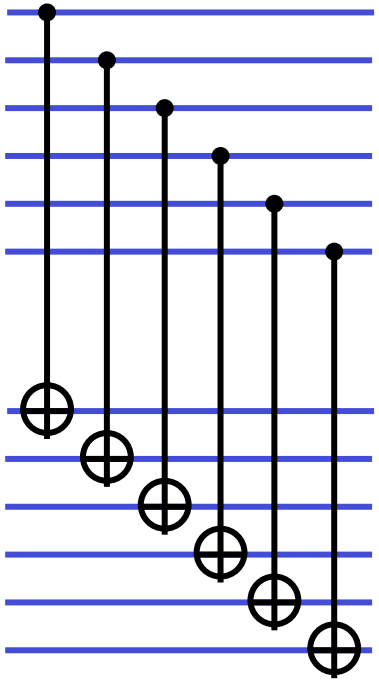
three qubit error

$$\exp[-i\mathbf{XXX}(t+\Delta t)] \approx -i\mathbf{XXX}\,[\mathbf{I} - i\Delta t(\mathbf{XII} + \mathbf{IXI} + \mathbf{IIX})]$$

single qubit errors

Single faulty component produces only one error per block of the code

Holy Grail of Fault Tolerant Computation



Block 1

Transverse operation

Block 2

Single error in Block 1 becomes error in Block 1 and error in Block 2

Beautiful Details

Fault tolerant correction procedures
(Fault tolerant measurements of stabilizer elements)

Fault tolerant measurement procedures

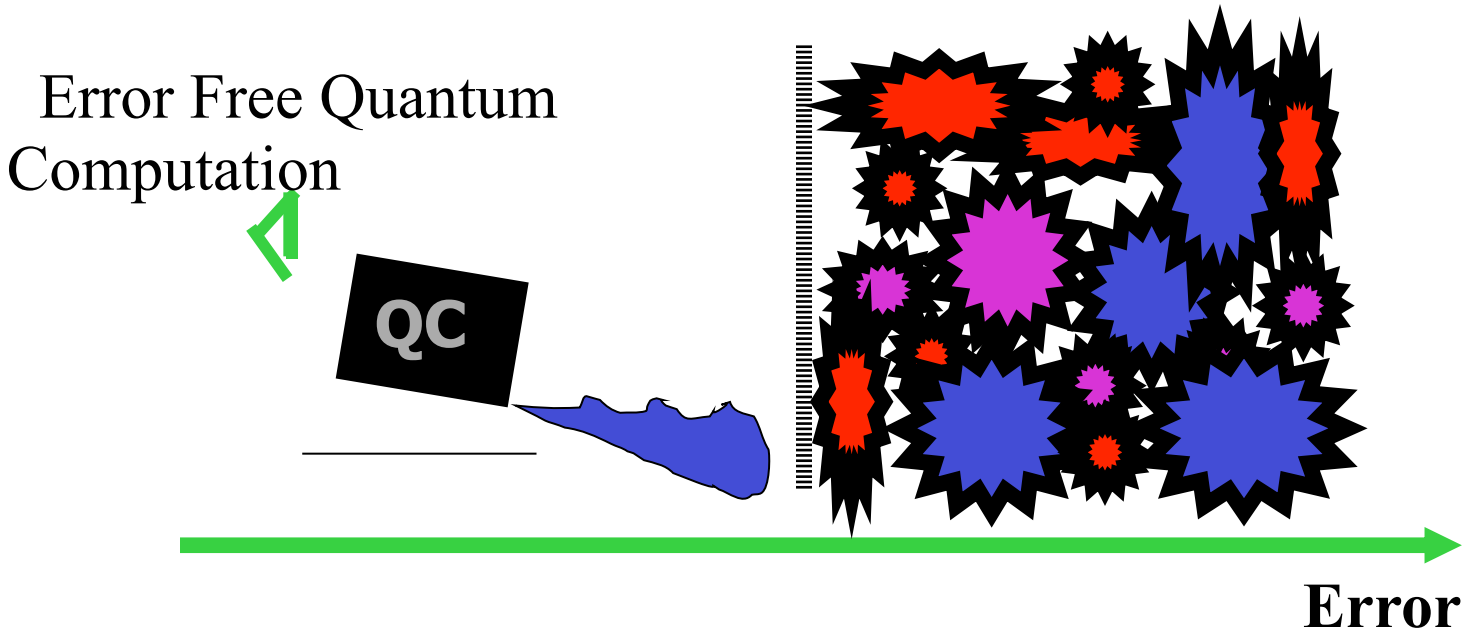
Fault tolerant preparation procedures

Fault tolerant universality procedures
(Clifford group elements often nice here!)

Threshing Out a Threshold

Competition between errors and quantum error correction

Fault tolerant routines do not spread errors like the plague



If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.

Some Rough Details

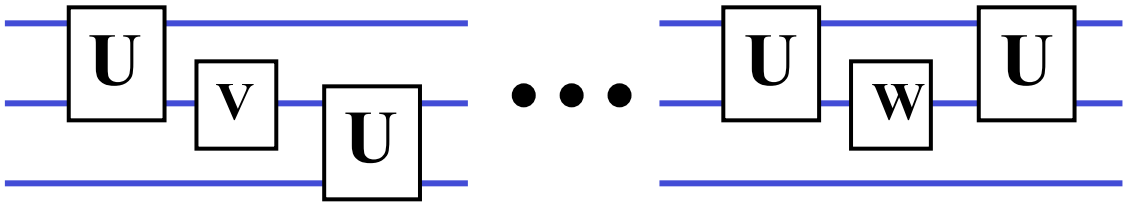
Failure probability for each level of encoding: $p \to cp^2$

$$p \to cp^2 \to c(cp^2)^2 \to \cdots \to \frac{(cp)^{2^k}}{c}$$

c: constant related to size of fault tolerant procedures

k levels of concatenation uses d^k gates.

Quantum algorithm with $p(n)$ gates:



Want algorithm with error probability ϵ: gates to accuracy $\frac{\epsilon}{p(n)}$

$$\frac{(cp)^{2^k}}{c} = \frac{\epsilon}{p(n)}$$

If $p < p_{thresh} = \frac{1}{c}$, then k levels rapidly get us below this threshold

Profound Truth?

Fight entanglement with entanglement.